EXISTING 12-bit RCA

1)EAL1

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	7.5	15.5	31.5	63.5	127.5
NORMALISED MEAN ERROR DISTANCE (NMED)	0.000915639	0.001892320	0.003845684	0.007752411	0.01556586
ERROR RATE (total no. of cases =16777216)	11468800	12795904	13791232	14537728	15097600
ACCURACY	31.64 %	23.7304%	17.7978%	13.3483%	10.01129%

AFA2 :- SUM = A^B^Cin

Cout= Cin

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	27.20 db	30.43 db	23.06 db	24.03 db	17.09 db	17.30 db	12.78 db	12.86 db	6.13 db	6.18 db
Laplacian kernel	6.44 db	6.37 db	13.28 db	6.51 db	12.78 db	6.38 db	11.28 db	6.20 db	11.24 db	6.33 db	11.50 db
Sobel kernel	6.52 db	6.59 db	22.48 db	6.68 db	19.24 db	6.43 db	14.31 db	6.17 db	13.32 db	6.38 db	13.34 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	63	110.124	7.886	2161.933	659.807	2821.740	513 ps
5-bit approx	49	87.552	6.090	1638.308	513.484	2151.792	441 ps
6-bit approx	43	78.660	5.430	1385.490	442.095	1827.585	369 ps
7-bit approx	37	69.768	4.769	1134.485	371.155	1505.641	298ps
8-bit approx	31	60.876	4.106	891.245	302.209	1193.454	226 ps

2) EAL2

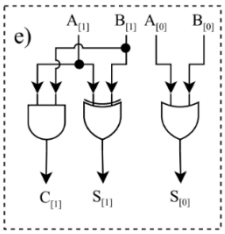
2-AxB1 AND 2-AxB2 adder logic

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	2.4375	NA	9.9765625	NA	40.053
NORMALISED MEAN ERROR DISTANCE (NMED)	0.00029758	NA	0.001217990	NA	0.00488996
ERROR RATE (total no. of cases =16777216)	13631488	NA	15597568	NA	16334848
ACCURACY	18.75 %	NA	7.0312%	NA	2.6367%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	29.07 db	36.40 db	NA	NA	23.74 db	24.90 db	NA	NA	13.71 db	13.81 db
Laplacian kernel	6.44 db	6.28 db	13.42 db	NA	NA	6.32 db	12.78 db	NA	NA	5.89 db	11.67 db
Sobel kernel	6.52 db	6.48 db	24.05 db	NA	NA	6.46 db	19.70 db	NA	NA	5.60 db	13.82 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	59	98.496	6.991	2077.555	629.264	2706.819	584 ps
5-bit approx	NA	NA	NA	NA	NA	NA	NA
6-bit approx	46	76.608	5.422	1534.507	471.134	2005.642	441 ps
7-bit approx	NA	NA	NA	NA	NA	NA	NA
8-bit approx	33	54.720	3.851	991.999	313.075	1305.074	298 ps

3) EAL3

LXOA adder logic 

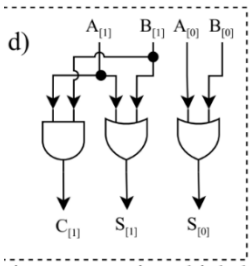
Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	1.75	3.75	7.75	15.75	31.75
NORMALISED MEAN ERROR DISTANCE (NMED)	0.000213649	0.000457819	0.000946160	0.0019228421	0.0038762
ERROR RATE (total no. of cases =16777216)	9699328	11468800	12795904	13791232	14537728
ACCURACY	42.1875%	31.64%	23.73%	17.7978%	13.33483%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	29.07 db	36.39 db	27.59 db	31.32 db	23.72 db	24.88 db	17.77 db	18.01 db	13.42 db	13.51 db

Laplacian kernel	6.44 db	6.25 db	13.44 db	6.27 db	24.01 db	6.34 db	13.02 db	6.32 db	12.13 db	6.25 db	12.34 db
Sobel kernel	6.52 db	6.46 db	24.17 db	6.48 db	23.10 db	6.46 db	20.45 db	6.25 db	16.13 db	5.93 db	14.80 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	61	101.232	7.159	2105.898	641.364	2747.262	584 ps
5-bit approx	55	90.972	6.417	1841.832	565.427	2407.259	513ps
6-bit approx	49	80.712	5.675	1577.022	489.284	2066.307	441 ps
7-bit approx	43	70.452	4.932	1311.947	413.055	1725.002	369ps
8-bit approx	37	60.192	4.188	1048.685	337.275	1385.961	298 ps

4) EAL4



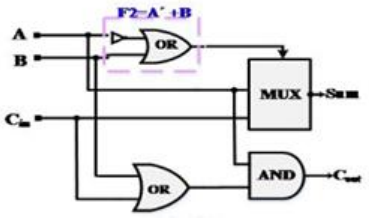
Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	2.875	5.875	11.875	23.875	47.875
NORMALISED MEAN ERROR DISTANCE (NMED)	0.00035099	0.00071725	0.0014497619	0.0029147	0.0058448

ERROR RATE (total no. of cases =16777216)	11468800	12795904	13791232	14537728	15097600
ACCURACY	31.6406%	23.73%	17.797%	13.3483%	10.011%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	29.50 db	39.76 db	28.58 db	34.25 db	24.87 db	26.60 db	17.32 db	17.55 db	13.32 db	13.24 db
Laplacian kernel	6.44 db	6.34 db	13.31 db	6.30 db	12.98 db	6.21 db	12.20 db	5.99 db	10.26 db	5.30 db	12.25 db
Sobel kernel	6.52 db	6.51 db	23.13 db	6.44 db	20.35 db	6.27 db	16.37 db	5.69 db	10.93 db	5.85 db	13.52 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	61	99.864	7.076	2093.641	638.944	2732.585	584 ps
5-bit approx	55	89.604	6.334	1829.57	563.007	2392.582	513 ps
6-bit approx	49	79.344	5.592	1564.765	486.864	2051.630	441 ps
7-bit approx	43	69.084	4.849	1299.690	410.635	1710.325	369 ps
8-bit approx	37	58.824	4.105	1036.428	334.855	1371.824	298 ps

5) EAL5



Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	4.46093	9.125	18.395	36.890	73.8510
NORMALISED MEAN ERROR DISTANCE (NMED)	0.00054461	0.001114027	0.002245799	0.004503844	0.009016124
ERROR RATE (total no. of cases =16777216)	13172736	14417920	15233024	15766528	16115712
ACCURACY	21.484%	14.06%	9.204%	6.02416%	3.9428%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	26.82 db	29.65 db	22.41 db	23.23 db	16.29 db	16.46 db	11.86 db	11.92 db	5.15 db	5.19 db

Laplacian kernel	6.44 db	6.26 db	13.23 db	6.20 db	12.56 db	5.87 db	11.07 db	5.39 db	10.77 db	5.15 db	10.02 db
Sobel kernel	6.52 db	6.48 db	22.62 db	6.40 db	19.41 db	5.98 db	14.85 db	5.30 db	13.24 db	5.15 db	11.64 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	89	136.8	9.725	2539.334	524.552	3363.886	752 ps
5-bit approx	91	136.116	9.610	2421.934	809.907	3231.841	726 ps
6-bit approx	93	135.432	9.495	2304.350	795.280	3099.630	700 ps
7-bit approx	95	134.748	9.380	2186.242	780.592	2966.834	674ps
8-bit approx	97	134.064	9.265	2068.967	766.174	2835.141	648 ps

6) EAL6

SUM =A^B^C AND Cout=B

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	5.515625	11.1367	22.3525	44.76440	89.573303
NORMALISED MEAN ERROR DISTANCE (NMED)	0.00067337	0.00135962	0.0027289	0.00546507	0.0109355760
ERROR RATE (total no. of cases =16777216)	11468800	12795904	13791232	14537728	15097600
ACCURACY	31.6406%	23.7304%	17.7978%	13.34838%	10.01129%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	29.44 db	39.23 db	28.16 db	32.91 db	23.99 db	25.36 db	17.26 db	17.48 db	13.12 db	13.05 db
Laplacian kernel	6.44 db	6.48 db	13.23 db	6.54 db	12.75 db	6.60 db	11.78 db	6.31 db	9.51 db	6.31 db	10.40 db
Sobel kernel	6.52 db	6.62 db	22.52 db	6.67 db	19.21 db	6.63 db	15.04 db	5.87 db	9.46 db	6.17 db	11.59 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	63	110.124	7.865	2269.458	685.839	2955.297	568 ps
5-bit approx	58	103.284	7.399	2067.567	627.933	2695.500	496ps
6-bit approx	53	96.444	6.392	1865.886	570.080	2435.966	424 ps
7-bit approx	48	89.604	6.465	1665.045	512.447	2177.492	352ps
8-bit approx	43	82.764	5.998	1466.643	455.450	1922.093	281 ps

7) EAL7

SUM=A`B`+C`

COUT =AB+CA

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	6.125	12.26953125	24.47949	48.844726	97.53948
NORMALISED MEAN ERROR DISTANCE (NMED)	0.000747777	0.001497928	0.002988584	0.005963218	0.01190812
ERROR RATE (total no. of cases =16777216)	15728640	16252928	16515072	16646144	16711680
ACCURACY	6.25%	3.125%	1.5625%	0.78125%	0.390625%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	29.44 db	39.26 db	28.23 db	33.27 db	24.41 db	25.79 db	21.68 db	22.46 db	12.10 db	12.31 db

Laplacian kernel	6.44 db	6.42 db	13.32 db	6.60 db	12.89 db	6.67 db	11.56 db	6.57 db	10.81 db	6.60 db	9.80 db
Sobel kernel	6.52 db	6.65 db	22.53 db	6.82 db	19.62 db	6.74 db	15.01 db	6.64 db	13.21 db	6.88 db	12.00 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	72	119.016	8.184	2410.953	797.459	3208.412	739 ps
5-bit approx	69	113.886	7.746	2220.358	761.668	2982.026	710 ps
6-bit approx	66	108.756	7.307	2029.307	725.750	2755.057	680 ps
7-bit approx	63	103.626	6.868	1838.430	689.868	2528.298	651ps
8-bit approx	60	98.496	6.428	1648..717	654.279	2302.996	622 ps

8) EAL8

SUM=B+Cin

Cout=AB+AC

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	5.3125	10.65625	21.328125	42.6640625	85.33203125
NORMALISED MEAN ERROR DISTANCE (NMED)	0.000648577	0.0013009705	0.002603848	0.00520865	0.01041777
ERROR RATE (total no. of cases =16777216)	15728640	16252928	16515072	16646144	16711680

ACCURACY	6.25%	3.125%	1.5625%	0.78125%	0.390625%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	29.11 db	37.01 db	27.25 db	30.81 db	22.86 db	23.87 db	14.64 db	14.81 db	8.47 db	8.53 db
Laplacian kernel	6.44 db	6.56 db	13.30 db	6.69 db	12.96 db	6.82 db	11.84 db	5.87 db	8.98 db	4.71 db	8.20 db
Sobel kernel	6.52 db	6.66 db	22.50 db	6.74 db	19.69 db	6.81 db	15.92 db	6.46 db	12.00 db	5.02 db	9.35 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	63	102.6	7.297	2207.067	663.639	2870.706	713 ps
5-bit approx	58	93.708	6.658	1982.390	597.215	2579.604	684 ps
6-bit approx	53	84.816	6.019	1756.973	530.589	2287.563	655 ps
7-bit approx	48	75.924	5.379	1531.526	463.944	1995.469	626ps

8-bit approx	43	67.032	4.739	1307.804	397.735	1705.639	597 ps

9) EAL9

SUM =B + Cin

Cout= A

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	5.3125	10.65625	21.328125	42.6640625	85.33203125
NORMALISED MEAN ERROR DISTANCE (NMED)	0.000648577	0.0013009705	0.002603848	0.00520865	0.01041777
ERROR RATE (total no. of cases =16777216)	15728640	16252928	16515072	16646144	16711680
ACCURACY	6.25%	3.125%	1.5625%	0.78125%	0.390625%

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel

Box kernel	29.10 db	27.98 db	32.52 db	24.88 db	26.59 db	20.07 db	20.79 db	13.23 db	13.34 db	7.20 db	7.26 db
Laplacian kernel	6.44 db	6.30 db	13.39 db	6.31 db	13.15 db	6.32 db	12.51 db	6.08 db	11.37 db	5.15 db	9.72 db
Sobel kernel	6.52 db	6.53 db	23.42 db	6.57 db	21.23 db	6.54 db	17.62 db	6.23 db	15.39 db	5.42 db	12.41 db

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	59	97.128	6.872	2061.939	622.314	2684.253	568 ps
5-bit approx	53	86.868	6.130	1799.685	546.863	2346.548	496ps
6-bit approx	47	76.608	5.388	1537.641	471.465	2009.107	424 ps
7-bit approx	41	66.348	4.646	1276.437	396.287	1672.724	352ps
8-bit approx	35	56.088	3.903	1017.672	321.745	1339.417	281 ps